MITSUBISHI GT1000PN RSX GASOLINE ENGINE

Test Report No	:	2014 - 04
Date of Test	:	January 21, 2014
Test Valid Until	:	January 21, 2019
Brand	:	MITSUBISHI
Model	:	GT1000PN RSX
Serial No.	:	WH0004511
Type	:	Four stroke, air-cooled, OHV, single inclined cylinder gasoline engine
Manufacturer	:	Made in Japan
Test Requested by	:	SEACOM INC. 721 Aurora Blvd., New Manila, Quezon City
Machine Classification	:	Commercial

SUMMARY

The MITSUBISHI GT1000PN RSX gasoline engine was tested to establish its performance. It was tested at the full throttle speed setting pre-set by the manufacturer/ test applicant.

The Maximum Power developed by the engine was 5.47 kW at 3602 rpm engine Speed. At Maximum Power, the Torque on the output shaft of the engine was 1.48 kg-m with Fuel Consumption Rate of 3.11 L/h. The Noise Level measured at this condition was 89.9 db(A). The Maximum Torque developed was 1.61 kg-m at an Engine Shaft Speed of 2907 rpm.

For the Continuous running test, the engine was set at the Maximum Power it can attain in its Rated Shaft Speed. The Average Output Power for the five-hour test was 5.20 kW at 3622 rpm Engine Shaft Speed.

The engine was easy to start. It only took a single attempt to start the engine using rope-recoil in either cold or hot starting.

SCOPE OF TEST

The engine was tested to establish its performance characteristics. The test was conducted on January 21, 2014 at the AMTEC Test Laboratory. It was evaluated on the following items:

a. Suitability of operator's manual;
b. Starting test;
c. Varying load performance test; and
d. Continuous running test

METHOD OF TEST

The engine was tested based on the Philippine Agricultural Engineering Standards, PAES 117:2000 Agricultural Machinery – Small Engines – Method of Test.

DESCRIPTION OF THE ENGINE

The MITSUBISHI GT1000PN RSX gasoline engine (**Figure 1**) had a Rated Maximum Output Power of 7.46 kW (10.0 Hp) at Rated Engine Shaft Speed of 3600 rpm, based on the owner's manual provided. It was a four-stroke, air-cooled, OHV, single inclined cylinder gasoline engine. Fuel feed system was gravity type with the fuel tank located on top of the engine. It utilized a semi-dry polyurethane foam moistened with oil. Lubrication system was splash type. Cooling was done by air forced by a fan/ flywheel encased in a metal shroud. It was started using rope-recoil starting system. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the engine.

Figure 1. MITSUBISHI GT1000PN RSX Gasoline Engine

ENGINE SPECIFICATIONS

Table 1. Detailed engine specifications of the MITSUBISHI GT1000PN RSX gasoline engine

1.	Dimensions and Weight		
	1.1 Overall length (mm)	:	429
	1.2 Overall width (mm)	:	392
	1.3 Overall height (mm)	:	427
	1.4 Weight (kg)	:	31.0
2.	Engine specifications (Based on Plate rating/operating manual and ocular inspection of the engine)		
	2.1 Brand	:	MITSUBISHI
	2.2 Model	:	GT1000PN RSX
	2.3 Serial No.	:	WH0004511
	2.4 Make	:	JAPAN
	2.5 Type	:	Four stroke, air-cooled, OHV, single inclined cylinder gasoline engine
	2.6 Rated Output, kW (based on plate rating and operator's manual)		
	Maximum Power	:	7.46 @ 3600 rpm
	Continuous Power	:	5.50 @ 3600 rpm
	2.7 Bore x stroke (mm)	:	80 x 59
	2.8 Total displacement volume (cc)	:	389
	2.9 Fuel system		
	2.9.1 Type of fuel feed	:	Gravity
	2.9.2 Fuel used	:	Gasoline
	2.10 Cooling system	:	Air-cooled
	2.11 Lubricating system	:	Splash type
	2.12 Starting system	:	Rope-recoil
	2.13 Air cleaner	:	Semi-dry (polyurethane)
	2.14 Exhaust system	:	Muffler

OPERATOR'S MANUAL

An owner's manual applicable for MITSUBISHI GT1000 and GT1300 gasoline engines was included in the engine that was submitted. The manual was presented in six different languages including English. It contained the following: Safe Use, Component Names, Fuel and Engine Oil, Pre-Operation Engine Inspection, Starting and Stopping the Engine, Care of the Engine, Inspection Precaution Items, Inspections and Maintenance, Long Term Storage, and Trouble Shooting.

RESULTS AND DISCUSSION

The MITSUBISHI GT1000PN RSX gasoline engine was tested to establish its performance. The test was conducted by directly coupling the power output shaft of the engine to the electric dynamometer. **Figure 2** shows the test set-up.

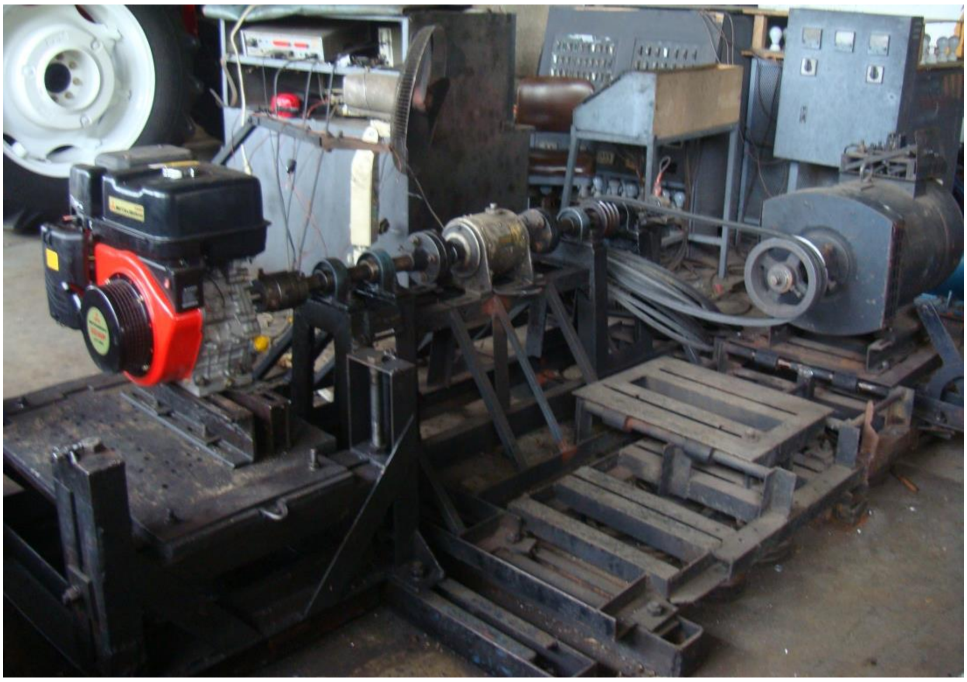

Figure 2. Test set-up of the MITSUBISHI GT1000PN RSX Gasoline Engine

A. STARTING TEST

Table 2 shows the result of the Starting test. The engine was started using the recommended starting procedure by the manufacturer. For cold starting, the engine was not operated for at least 12 hours and then started, while for hot starting, the engine was operated for one (1) hour and then stopped and re-started. The number of attempts needed to start the engine was recorded in both tests.

During cold and hot starting, it only took a single attempt to start the engine. In general, the engine was easy to start.

STARTING TEST

Engine on Test : MITSUBISHI Gasoliine Engine
Model : GT1000PN RSX
Date of Test : January 21, 2014

Table 2. Results of starting test

ITEMS	COLD STARTING	HOT STARTING
1. Ambient temperature (^oC)	24.7	
2. Relative Humidity (%)	75.0	
3. Engine oil temperature (^oC)	24.7	89.0
4. No. of attempts to start the engine	1	1
5. Ease of starting	Easy	Easy

B. VARYING LOAD PERFORMANCE TEST

The engine mounted on the electric dynamometer was tested for varying load performance. Data were obtained at the recommended maximum no-load speed setting (Pre-set) of the engine and at prevailing ambient conditions during the time of test using locally available gasoline fuel. **Table 3** and **Figure 3** show the results of the varying load performance test.

OUTPUT POWER. Based on the results of test, the Maximum Power developed by the engine, was 5.47 kW at 3602 rpm engine shaft speed. At Maximum Power, the Torque on the output shaft of the engine was 1.48 kg-m with Fuel Consumption Rate of 3.11 L/h. The Maximum Torque developed was 1.61 kg-m at an Engine Shaft Speed of 2907 rpm. The Actual Maximum Power was 73.3% of the Rated Maximum Power based on plate rating of the engine.

FUEL CONSUMPTION. The Fuel Consumption of the engine was obtained using ONO SOKKI Digital Fuel Consumption Meter Model: FC-014/024 coupled to an ONO SOKKI Burette Stand Model: PP-523. The Maximum Fuel Consumption Rate of the engine was 3.14 L/h. It occurred at 5.27 kW Output Power of the engine and at 3802 rpm Engine Shaft Speed. The Specific Fuel Consumption at this point was 432.4 g/kW-h.

TEMPERATURES. Temperatures were measured using Type J thermocouples attached to a YOKOGAWA Digital Thermometer, Type 2575 coupled to YOKOGAWA multi-channel switch, Type 2815. The Maximum Temperature of the engine oil was 109.5^{o}C which occurred at 5.24 kW Output Power of the engine and at 3702 rpm Engine Shaft Speed. For the exhaust air temperature at the outlet of the muffler, the Maximum Temperature was 691.5^{o}C which occurred at 4.61 kW Output Power and at 3998 rpm Engine Shaft Speed.

NOISE LEVEL. Noise Level was the quantified degree of sound emitted by the engine during operation. It was measured using a SIMPSON Sound Level Meter Model: 884-2. The Noise Level of the engine was measured at 7.5 meters away from the muffler, 1.2 meters high from the ground and 45^{o} from the engine shaft. The Maximum Noise Level measured was 94.8 db(A).

OBSERVATIONS. No breakdown and other malfunctioning occurred during the test.

VARYING LOAD PERFORMANCE TEST

Engine on Test : MITSUBISHI Gasoline Engine

Model : GT1000PN RSX

Date of Test : January 21, 2014

Test Conditions

Ambient Temperature (°C) : 28.7
Relative humidity (%) : 93.6

Table 3. Results of varying load performance test[1]

SHAFT TORQUE (kg-m)	ENGINE SHAFT SPEED (rpm)	OUTPUT POWER (kW)	FUEL CONSUMP-TION (L/h)	SPECIFIC FUEL CONSUMP-TION (g/kW-h)	TEMPERATURE (°C)		NOISE LEVEL [DB(A)]
					ENGINE OIL	EXHAUST AIR	
0.72	4276	3.15	2.67	614.7	89.0	574.0	94.0
0.90	4192	3.88	2.72	508.5	93.5	616.0	94.8
1.08	4094	4.52	2.88	461.9	97.0	665.5	94.6
1.12	3998	4.61	2.97	466.5	100.0	691.5	93.8
1.26	3898	5.02	3.05	440.7	103.0	686.0	92.4
1.35	3802	5.27	3.14	432.4	108.0	683.0	91.9
1.38	3702	5.24	3.13	433.7	109.5	680.5	91.6
1.48	3602	5.47	3.11	412.2	108.5	670.0	89.9
1.52	3497	5.46	3.03	402.7	108.0	659.5	87.9
1.55	3402	5.41	2.99	400.5	105.5	648.5	87.0
1.60	3303	5.44	2.89	385.4	105.0	635.5	86.4
1.60	3201	5.25	2.88	398.4	103.5	616.0	86.2
1.57	3103	5.01	2.78	401.9	103.0	615.5	86.4
1.57	2995	4.81	2.68	403.6	102.0	598.5	85.8
1.61	2907	4.80	2.61	394.8	101.5	587.0	84.8
1.59	2805	4.57	2.50	397.0	100.5	574.5	84.2
1.58	2698	4.37	2.38	394.7	100.0	548.5	83.7
1.60	2604	4.29	2.36	398.9	100.0	545.0	83.2
1.58	2499	4.04	2.29	411.0	100.0	532.5	83.1
1.57	2404	3.88	2.17	405.8	99.5	523.0	82.9
1.58	2297	3.73	2.07	402.5	99.5	514.0	83.4
1.57	2198	3.54	1.97	402.5	99.0	501.5	82.8
1.52	2094	3.27	1.89	418.6	99.0	488.0	82.6
1.51	2001	3.09	1.79	419.8	99.0	472.5	82.1
1.53	1901	2.99	1.68	408.0	98.5	446.0	81.8
1.43	1804	2.66	1.61	438.4	98.5	425.0	81.4
1.44	1702	2.51	1.51	436.8	98.5	409.5	81.7
1.44	1598	2.36	1.42	436.0	98.0	403.0	81.8
1.36	1501	2.10	1.35	464.2	98.0	393.0	81.9

[1] Average of three test trials

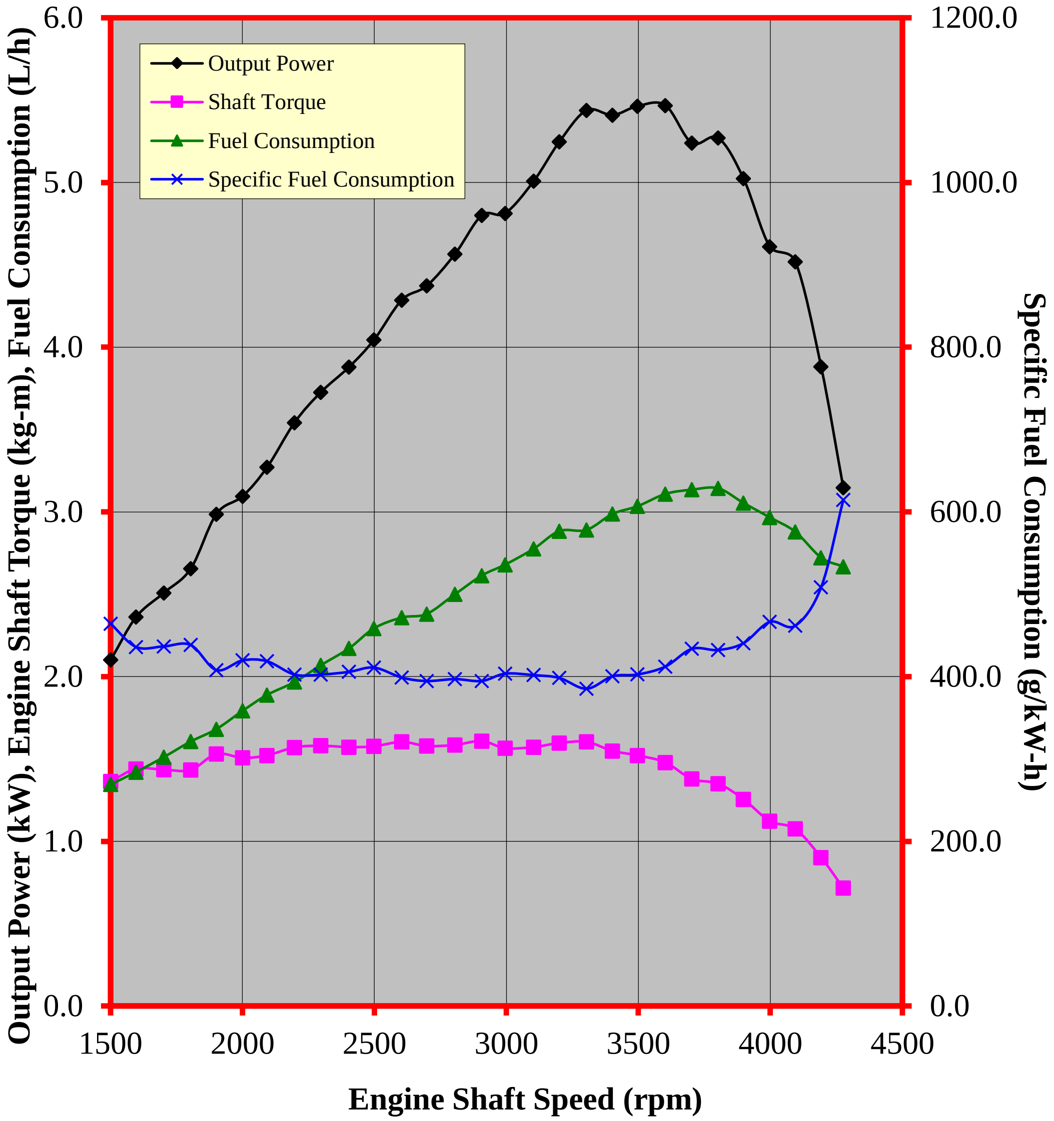

Figure 3. Performance characteristic curves of the MITSUBISHI GT1000PN RSX gasoline engine

C. CONTINUOUS RUNNING TEST

The engine mounted on the electric dynamometer was made to run continuously for five hours. Initially, the throttle was set at the no-load speed setting as in the varying load test. As per standard procedure, the engine should be set at the Rated Continuous Power specified by the manufacturer. However, if not specified, the engine should be set at 80% of the Rated Maximum Power. If the engine cannot attain the 80% of the Rated Maximum Power specified by the manufacturer, it should be set at the Maximum Power it can attain at Rated Speed. As such, the engine was set at the Maximum Power it can attain in its Rated Shaft Speed. Data were gathered every thirty minutes. No breakdown occurred.

The Average Output Power obtained after the five-hour period was 5.2 kW at 3622 rpm Engine Shaft Speed. This was 69.7% of the Rated Maximum Output Power. Results of test are shown in **Table 4** and plotted in **Figure 4**.

CONTINUOUS RUNNING TEST

Engine on Test	:	MITSUBISHI Gasoline Engine
Model	:	GT1000PN RSX
Date of Test	:	January 21, 2014
Duration of Test (h)	:	5

Test Conditions

Ambient Temperature (°C)	:	26.1
Relative humidity (%)	:	74.9

Table 4. Results of continuous running test at 69.7 % of the Rated Maximum Power.

Items Measured	Average values
Engine Speed (rpm)	3622
Torque (kg-m)	1.40
Output Power (kW)	5.20
Fuel Consumption (L/h)	3.50
Specific Fuel Consumption (g/kW-h)	488.0
Temperature (°C) :	
a) Engine Oil	103.5
b) Exhaust Gas	504.9
Noise Level [db(A)]	90.4

OBSERVATIONS/REMARKS: No breakdown occurred.

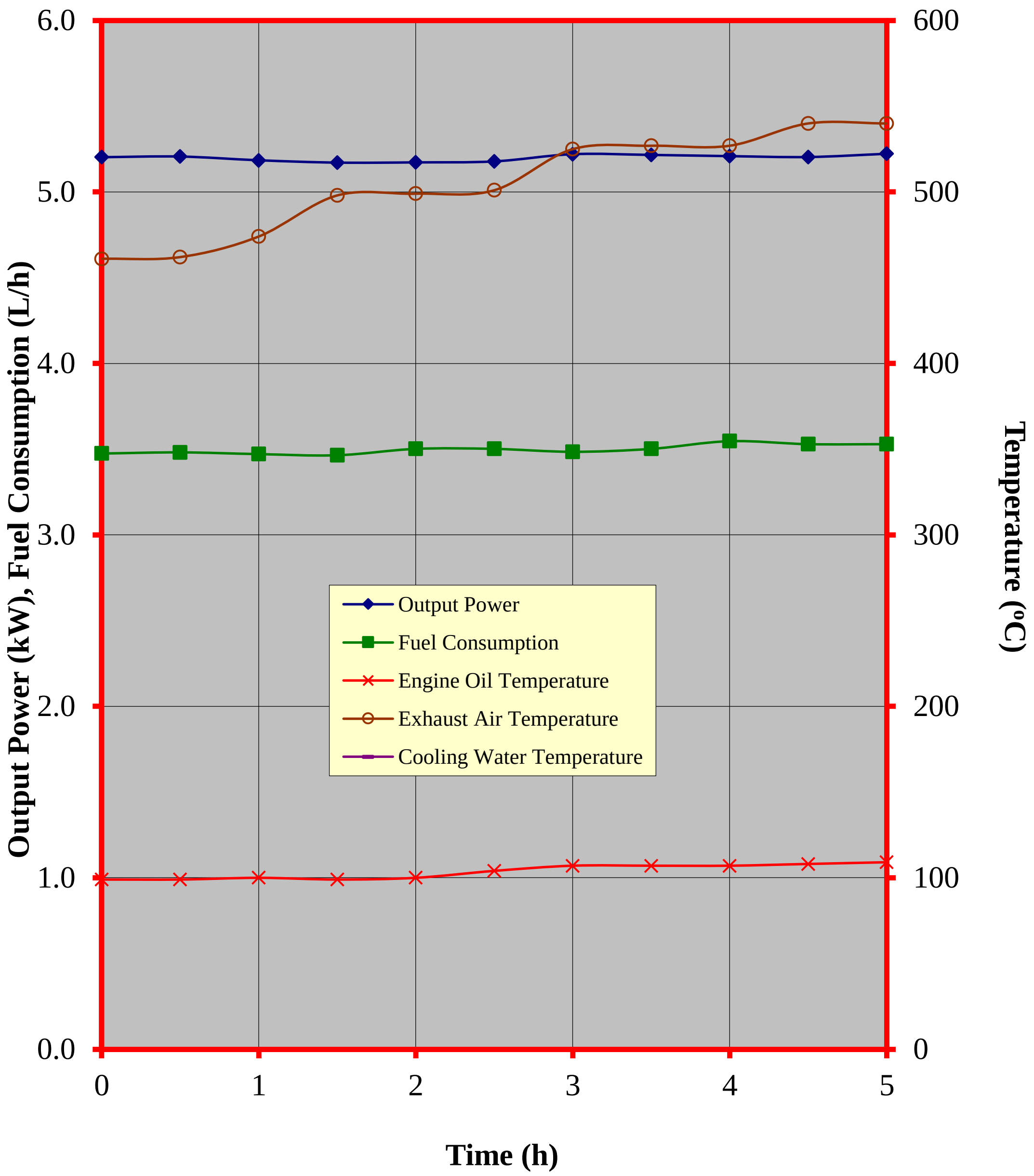

Figure 4. Performance characteristic curves during the continuous running test of the MITSUBISHI GT1000PN RSX gasoline engine

DATA COMPUTATION

ANNEX. Formula used during calculation and testing

1. Varying Load Test

The engine was mounted on a dynamometer. The engine throttle was set at the maximum setting recommended by the manufacturer/test applicant. Data were gathered at different engine speeds starting from the maximum no-load speed down to the lowest speed. (The engine speed should not be lower than 600 rpm.) Data such as shaft torque; shaft speed; fuel consumption; temperatures of engine oil, exhaust air, and cooling water; vibration level at x, y, and z axis; and noise level were gathered at every speed setting. The following were computed:

a. Output Power (kW), OP - The actual power at the output shaft of the engine is as given by equation 1.

$$OP = \frac{T \text{ X } N}{974} \quad , \text{ equation 1}$$

where:

$$\begin{aligned} OP &= \text{Output Power, kW} \\ T &= \text{Torque, kg - m} \\ N &= \text{Shaft Speed, rpm} \end{aligned}$$

$$\text{Percent power} = \frac{\text{Actual maximum power (kW) on test}}{\text{Rated maximum power (kW) found on the plate/brochure}} \text{ x } 100$$

b. Fuel Consumption (L/h) – It is the amount of fuel consumed by the engine per hour. This was obtained by volumetric method using ONO SOKKI Digital Fuel Consumption Meter (Model: FC-024) coupled to an ONO SOKKI Burette Stand (Model: PP-523).

Annex, Con't.

c. Specific Fuel Consumption (g/kW-h) - This is the amount of fuel consumed by the engine per unit of energy output. From test data, it was computed using equation 2.

$$SFC = \frac{FC \text{ X } D}{OP} \quad , \qquad \text{equation 2}$$

where:

$$SFC = \text{Specific Fuel Consumption, g/kW - h}$$
$$FC = \text{Fuel Consumption, L/h}$$
$$D = \text{Density of Fuel used, g/L}$$

d. Temperature (^{o}C) - These are the temperatures of the engine at different locations. They were measured using YOKOGAWA Digital Thermometer (Model: 2575) with Multi-Channel switch.

d.1 Engine Oil- the temperature of the engine oil at the crankcase during operation

d.2 Exhaust Air- the temperature of exhaust gas at the outlet of the muffler during operation

d.3 Cooling Water- the temperature of the water at the condenser during operation (if water cooled)

e. Noise Level [db(A)]- The noise emitted by the engine 7.5 meters away from the muffler, 45^{O} with respect to the axis of the engine shaft and the height of the microphone was 1.2 meters from the ground. **Table 5** shows the permissible exposure time for a certain level of sound emitted by the engine.

Table 5. Permissible exposure time for different sound levels

DURATION OF EXPOSURE PER DAY (h)	SOUND LEVEL [db(A)]
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25 OR LESS	115

Based on " General Industry Guide for applying Safety and Health Standards", 29CFR 1910 OSHA, U.S. Dept. of Labor, 1973

Annex, Con't.

2. Continuous Running Test

This is the durability test for the engine. The engine was set to the recommended continuous power rating and at rated shaft speed found on the engine plate and/or brochure and was operated continuously for five hours. Data (same as in varying load test) were gathered every 30 minutes. (If the engine has no recommended continuous power, it will be set at 80% of the rated maximum power and at rated shaft speed.) During the test, the shaft speed of the engine was not set lower than the rated. The average of the five-hour test data were reported as continuous test data of the engine. The average actual power of the engine divided by the rated maximum power found on the plate/brochure was reported as the percent continuous power attained by the engine.

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